This is an experiment using spatial loss in segmenting IsIc dataset

Spatial Focal Loss

To solve the main challenge of segmentation problem which is class imbalance as background pixels dominate most images, and lack of accrued labeled pixels so, will use focal loss + mask

During training, only a subset of pixels contribute to the loss (using masks).
Why not a standard Binary Cross Entropy?
Because it will treat easy and hard examples equally, and the loss would be dominated by background pixels, so focal loss would give less weights to easy examples classification (background) to influence the gradients values, and using the mask would adapt the network to deal with lack in segmented/labeled pixels

Focal Loss Overview

The idea is to down weight the influence of easy examples/ common class ex background pixels, that give the rare/hard examples ex lesion pixels the chance to have more influence to the loss value
Same BCE equation with extra 2 hyper parameters

- $\alpha \in [0,1] \rightarrow$ is a weight for each class, where the rare class is multiplied by a higher weight (α), and the common class is multiplied by a lower weight ($1-\alpha$), this increases the loss contribution of the rare class and decreases the contribution of the dominant class, so the model pays more attention to the rare class during training.

- Gamma → responsible for down weighting common/ easy examples to reduces the influence of easy examples on the loss function, resulting in more attention being paid to hard examples

To simulate sparse labeling, a binary supervision mask is applied, 1 → pixel contributes to loss and 0 → pixel ignored

Mask Sampling

To simulate partial supervision, a binary mask is generated to hide pixel values from the calculation in the loss function, where 1 → pixel contributes to loss and 0 → pixel ignored using 2 strategies.

1. Random Sampling:
   Randomly drops a percentage of pixels from the full mask, and it's useful for studying robustness under sparse supervision.

2. Class Balanced Sampling:
   Drops pixels separately from water class and background class, which is useful for preventing dominance of background class

Image Processing Pipeline

Loaded in **RGB format**, normalized for stable training, resized to a fixed resolution to reduce computational cost and ensure consistent input dimension
The downscaling is made separately where the image is interpolated using BICUBIC to preserve the details and sharper edges, and mask is interpolated using Nearest-exact to guarantee the closest match
Interpolate the decreased image using bicubic method to produce smoother downsampled images and preserve spatial details, besides using antialias to reduce high frequency artifacts when downsampling.
Finally normalize the image pixels in each channel to be in same range as the original pretrained images were trained for the encoder using imagenet dataset

Target Processing Pipeline

Loaded in **gray scale format as a single channel image**, resized to the same image fixed resolution but using a different Interpolation method.
Interpolated using NEAREST_EXACT method to prevent label interpolation, avoid fractional pixel values and preserve binary class integrity, then convert target pixels into 2 segments 0 for background, and 1 for lesion, target shape (H,W)

Model Configuration

Using U-Net with efficientnet encoder (ImageNet pretrained)
Training on Multi GPU: GPU T4 x2
Training using K-Fold Cross Validation strategy was adopted to ensure robust and unbiased model evaluation and more model generalization Number of folds: K = 3, each fold: 67% used for training and 33% used for validation

- Downsize images to (224,224)
- Batch Size 16 images
- Number of Epochs: 5
- Threshold: 0.5
- Loss Alpha: 0.25
- Loss Gamma: 2
- Adam optimizer
- Model Evaluation using Precision, Recall, F1 Score, IOU, and Dice

Model Evaluation Strategy

ISIC datasets often contain images with highly varying lesion areas:

- Some images contain large lesion regions
- Others contain only a few lesion pixels

To avoid bias evaluation, and ensure a fair and comprehensive assessment:

1. Global Pixel-Wise Evaluation
2. Per Image Spatial Evaluation

Global Pixel Wise Evaluation

This approach treats each pixel equally, regardless of which image it belongs to, It evaluates the model as a pixel-level binary classifier over the entire validation set.

This would reflect overall segmentation performance, suitable when viewing the task as large scale pixel classification and give strong signal about class imbalance behavior

All True Positives (TP), False Positives (FP), and False Negatives (FN) are aggregated across all pixels and all images

Then the following metrics are computed globally: **Precision, Recall and F1 Score**

Per Image Spatial Evaluation

This approach used to solve bias cases where images with large lesion areas dominate or images with small lesion regions barely influence the result

Using spatial metrics are computed per image, so for each image compute IOU and Dice then take the mean across all images

Intersection over union (IOU) and Dice are spatial metrics, they are helpful because they measure region overlap, evaluate segmentation quality and are sensitive to boundary alignment

Using both Strategies ensures the following:

- Fair evaluation across dataset imbalance
- Robust understanding of segmentation behavior
- Avoiding bias toward large lesion regions

Experiment 1 (Random Mask), Cross Validation Comparison

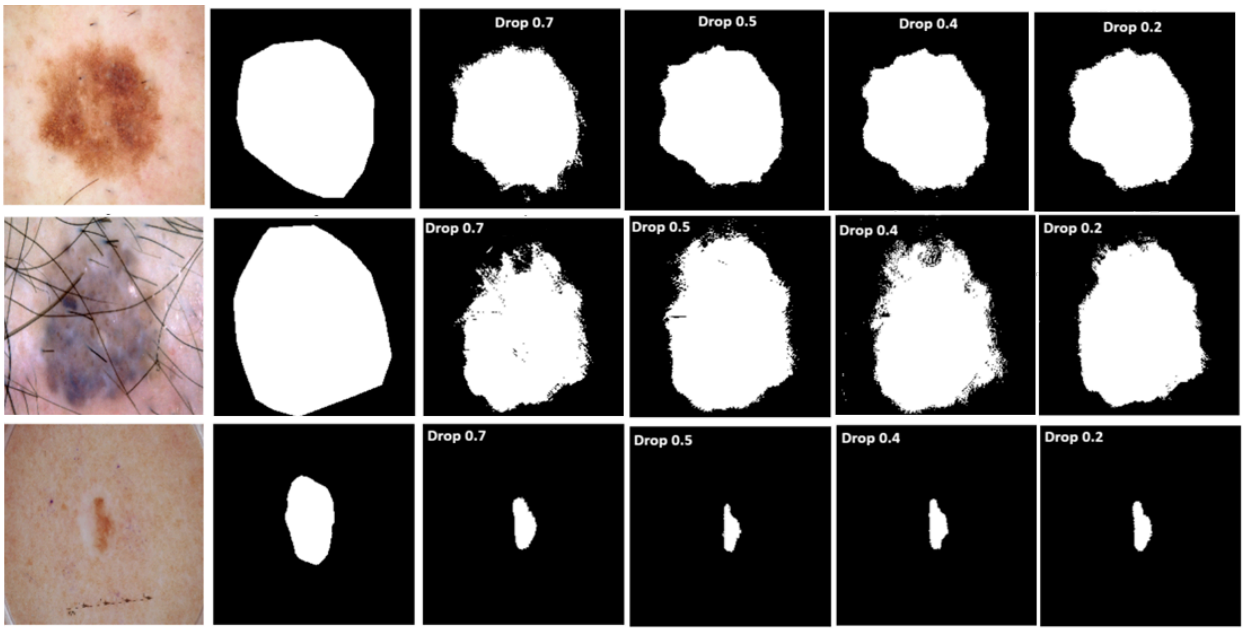
Drop	Precision	Recall	F1 Score	IoU	Dice
0.2	0.9590 ± 0.0057	0.8073 ± 0.0142	0.8765 ± 0.0067	0.7622 ± 0.0162	0.8455 ± 0.0122
0.4	0.9516 ± 0.0056	0.8122 ± 0.0025	0.8763 ± 0.0020	0.7628 ± 0.0027	0.8486 ± 0.0011
0.5	0.9574 ± 0.0092	0.8073 ± 0.0120	0.8759 ± 0.0041	0.7601 ± 0.0064	0.8436 ± 0.0043
0.7	0.9520 ± 0.0071	0.8204 ± 0.0137	0.8812 ± 0.0050	0.7689 ± 0.0074	0.8511 ± 0.0043

The conclusion:

- Since Precision/Recall/F1 scores are calculated over the whole validation set - per pixel, Comparing the model Precision(~95%) vs Recall (~81%), tell that the model is conservative when the model decide if this pixel is lesion it's 95% a lesion, but from all actual lesion pixels the model was able to catch 81% of them so the model prefer to be accurate rather than miss some actual pixels.
- Since IOU/ Dice are calculated per image, so the mean value of IOU (76%) tell that the model is able to cover 76% lesion area in the images on average
- Dice value in general is greater than the IOU value since it's double the value of intersected area, and Dice (84%) means that we have 84% matching between the ground truth/segmented image and the predicted/segmented image
- The std value for F1 score ex: drop 0.2, (0.0067) - over the pixels values in validation set - and the std values for the IOU (0.016)

and Dice(0.0122) tells that the predicted segmented per pixel/ per image are stable across the 3 folds which leads to consistent/ good segmentation quality output

- When comparing with drop 0.2 values, the average metrics values didn't have a noticeable change only Recall increased by 1% which leads detecting more actual pixels, however the std values over the 3 folds, have changes like in Recall, from 0.0142 to 0.0025
  F1 score from 0.0067 → 0.0020, IOU from 0.0162 → 0.0027 and Dice from 0.0122 → 0.0011
  This indicates that despite the increase of drop rate from 0.2 to 0.4 it yields more regularization; Made the model more consistency over folds without losing performance
  This suggests the model has become less sensitive to the specific pixel composition/ pattern of each fold's training split, indicating more robust feature learning.
  In another word the model isn't depending on specific pixel values like lesion texture and becomes more generalized on the shape of the lesion/ the context, less depending on pixel pattern and small details.
- Across all drop rates (0.2–0.7), the model maintained consistently high precision (95%) with gradually improving recall, indicating it becomes better at detecting actual lesion pixels without sacrificing prediction confidence.
- Contrary to expectations, drop=0.7 achieved the best IoU (0.7689) and Dice (0.8511); This can be attributed to two factors:
  - First, random pixel dropping preserves the overall lesion shape and context, allowing the model to learn robust spatial features even under heavy supervision sparsity.
  - Second, the relatively large training set (1,730 images per fold) provides sufficient diversity to generalize well. The narrow F1 range (0.876–0.881) across all experiments confirms stable and consistent performance.

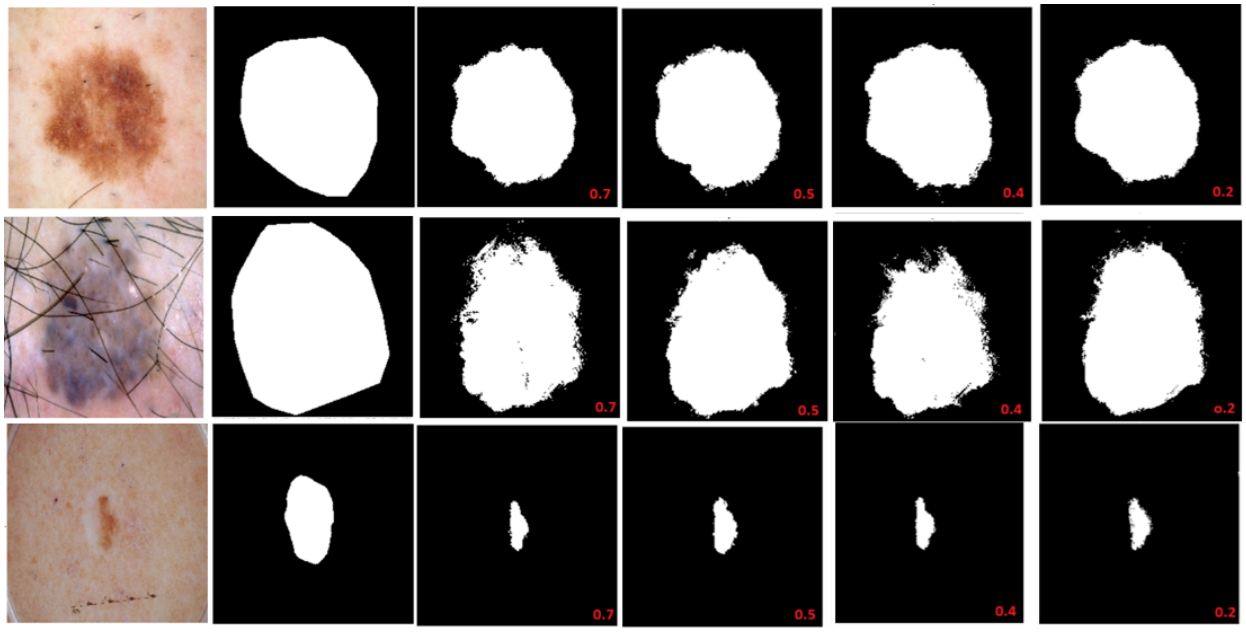


Visual inspection across three cases revealed two consistent challenges:

- First, all models were susceptible to hair artifacts, producing noisy predictions in images with dense hair, though this varied in severity across drop rates.
- Second, all models successfully detected the high-contrast dark core of lesions, but struggled with low-contrast lighter boundaries surrounding the lesion.
- Noting that, drop=0.7 showed better coverage of these lighter boundary regions, which aligns with its higher Recall value and explains its superior IoU score despite the added noise

Experiment 2 (Balanced Mask), Cross Validation Comparison

Drop Rate	Precision (Mean ± Std)	Recall (Mean ± Std)	F1 Score (Mean ± Std)	IoU (Mean ± Std)	Dice (Mean ± Std)
0.2	0.9535 ± 0.0126	0.7986 ± 0.0225	0.8689 ± 0.0103	0.7570 ± 0.0207	0.8402 ± 0.0175
0.4	0.9575 ± 0.0121	0.8000 ± 0.0226	0.8714 ± 0.0085	0.7575 ± 0.0159	0.8421 ± 0.0130
0.5	0.9560 ± 0.0047	0.8053 ± 0.0162	0.8741 ± 0.0076	0.7596 ± 0.0150	0.8423 ± 0.0143
0.7	0.9582 ± 0.0092	0.8041 ± 0.0132	0.8743 ± 0.0042	0.7526 ± 0.0096	0.8379 ± 0.0075



Final Conclusion:

Comparing the difference between random and balanced sampling strategies was minimal across all drop rates (IoU difference < 0.02). This suggests that on a large and diverse dataset like ISIC, the sampling strategy has limited impact on final performance, unlike smaller datasets where class-balanced supervision plays a more critical role.

The best overall performance was achieved with random sampling at drop=0.7, yielding IoU=0.7689 and Dice=0.8511, demonstrating that the model can perform robust lesion segmentation even under heavy sparse supervision.